5.4 Using Calculus to Graph Functions

Name _____ Period ____ Date _____

AP Calculus 5.4 · Student sheet

I can…

- find the vertical and horizontal asymptotes of a graph.
- use f′ to find where a graph rises and falls and its local extrema.
- use f″ to find concavity and inflection points, then sketch the graph.

Vocabulary · fill in the missing word

Vertical asymptote — the line x = _____ when f(x) → ±∞ as x → c

Horizontal asymptote — the line y = _____ when f(x) → L as x → ∞ or x → −∞

Part 1 · Build the picture

Sketch from the signs ($f(x) = x^4 - 2x^2$).

Interval	f′	f″	Shape
$(-\infty, -1)$	−	+	
$(-1, 0)$	+	+ then −	
$(0, 1)$	−	− then +	
$(1, \infty)$	+	+	

Part 2 · Notes and examples (fill in each note, then work the example)

Step 1: find the domain, the intercepts, and any _____.

Example 1

Find the asymptotes of $f(x) = \dfrac{2x - 4}{x + 1}$.

Step 2: use f′ to find where f increases and decreases and its local _____.

Example 2

$f(x) = x^4 - 2x^2$. Find the critical numbers and local extrema.

Step 3: use f″ to find the concavity and the _____ points.

Example 3

$f(x) = x^4 - 2x^2$. Find the inflection points.

Example 4

$f(x) = xe^x$. Find the local extremum and the inflection point.

Step 4: plot the key points and _____ the graph.

Example 5

Describe the graph of $f(x) = x^3 - 3x$.

Part 3 · Partner practice

Partner 1

Find the asymptotes of $f(x) = \dfrac{3x + 1}{x - 2}$.

Partner 2

(AP® Practice) How many inflection points does the graph of $f(x) = x^4 - 8x^2$ have?

(A) 2 (B) 0 (C) 1 (D) 3